

Twenty-year trends in excess costs of asthma

Background

In Canada, more than **3 million** people are living with asthma. We estimated the excess direct medical costs associated with asthma (difference between asthma and non-asthma groups) over a **20-year period**.

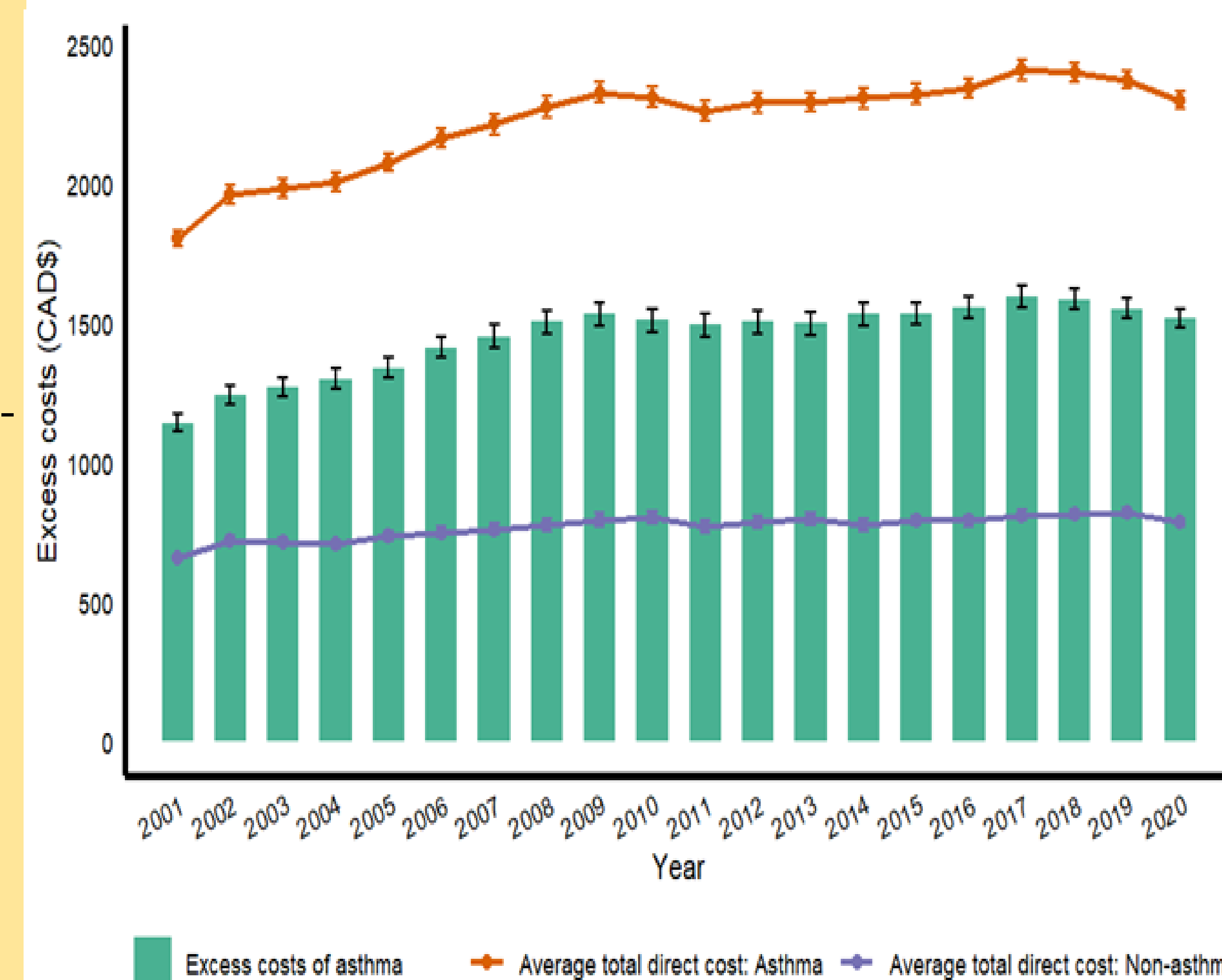
Methods

- Retrospective cohort study
- Individuals aged **5-55** years old
- Data: Administrative health data, British Columbia
- Matching: 1:2 propensity score matching with non-asthma population
- Excess Medical Costs categorized by:
 - Hospitalization
 - Outpatient
 - Medication
- Comorbidities: Assessed using ICD-10 codes
- Excess Costs: Difference between asthma and non-asthma group
- Costs expressed in 2020 CAD

Results

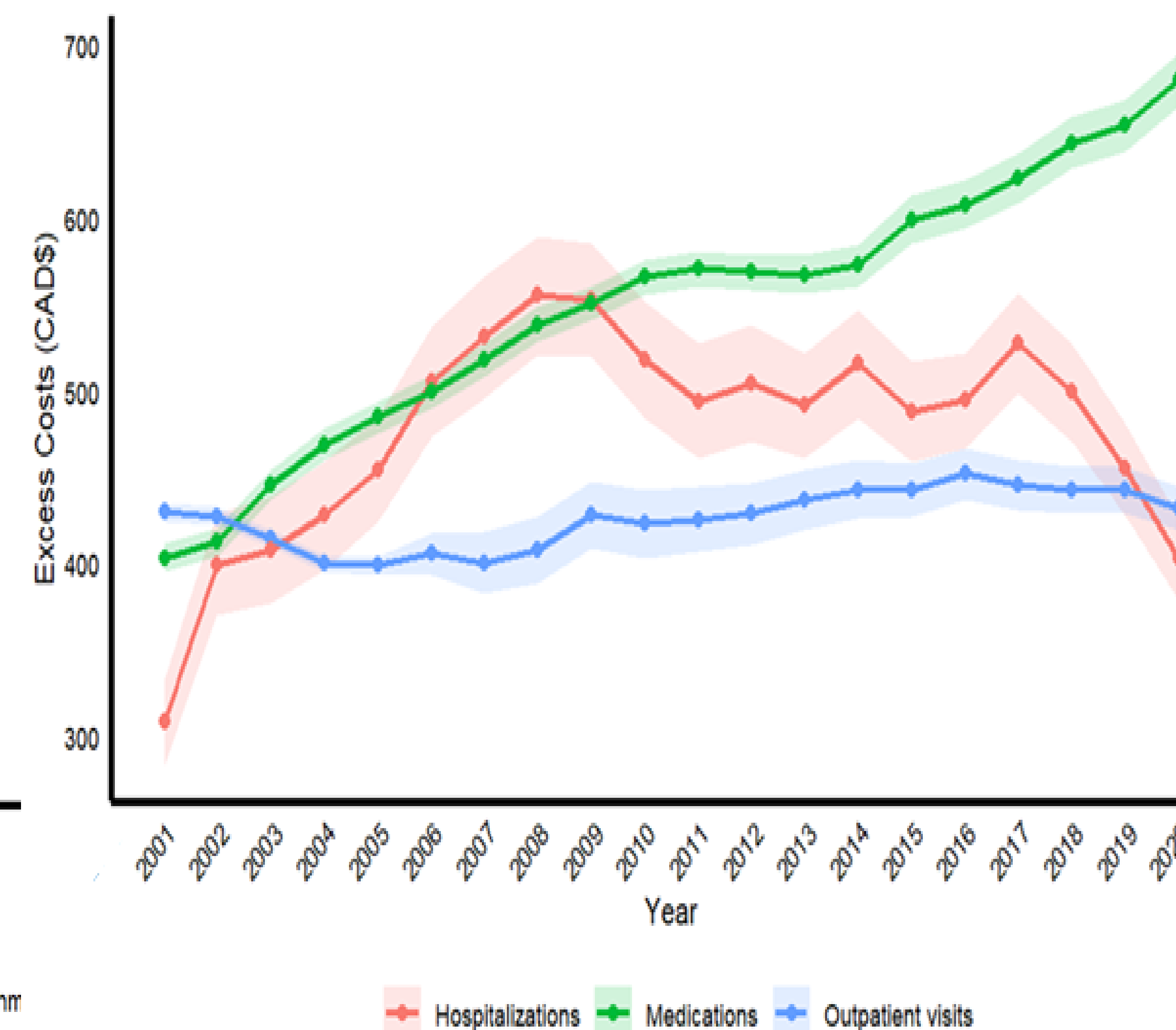
- 1,020,078 individuals
 - 510,039 in asthma (Prevalent cases)
 - 1,020,078 in non-asthma
- Mean age: 23.7 years
- Female: 52.4%

Patients with asthma incurred approximately **three times higher direct costs** than individuals without asthma

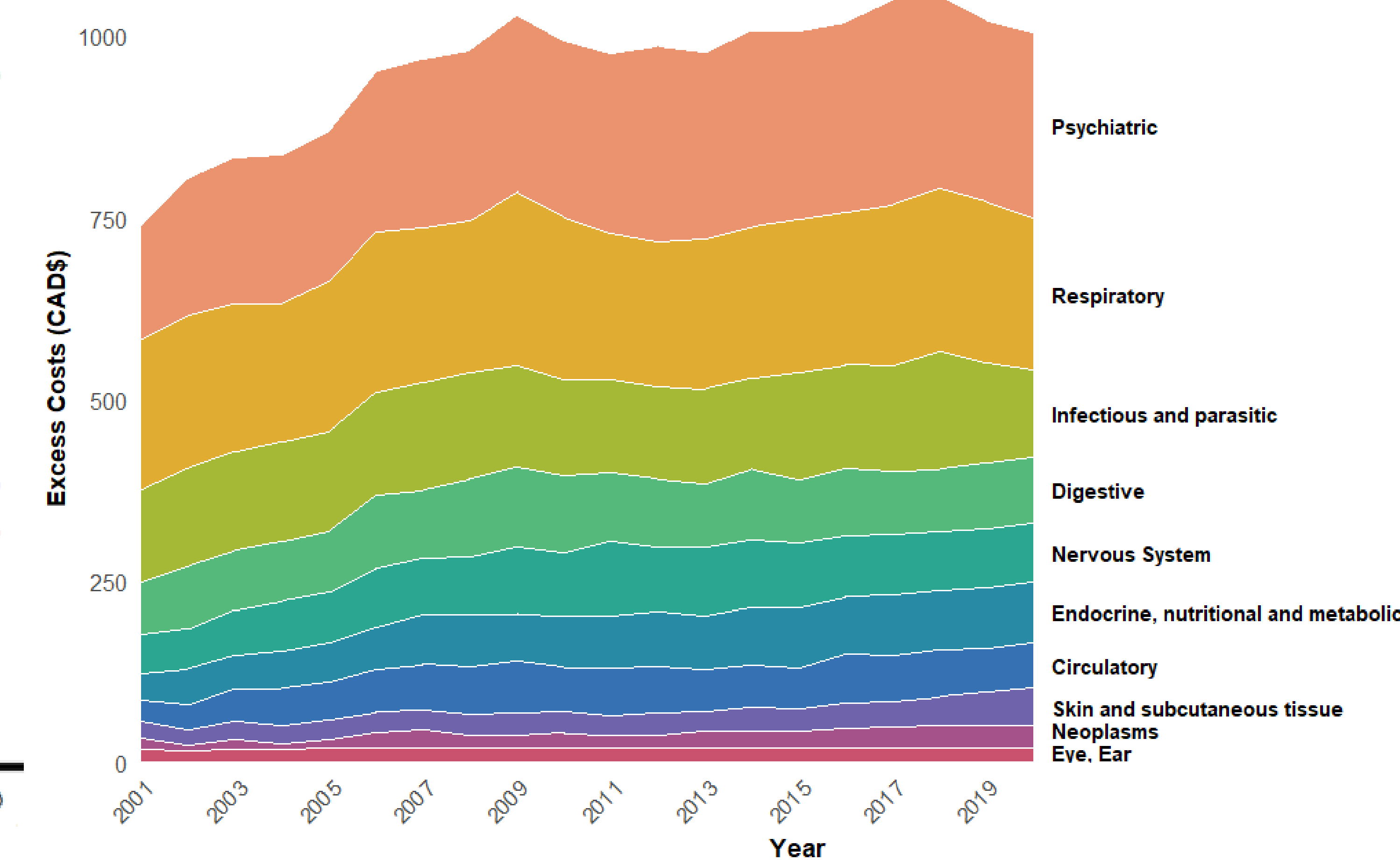


Excess costs for asthma patients were \$1,472/PY.

Excess costs increased by 33% over the 20-year study period



Medications accounted for the largest share (\$562/PY), followed by hospitalizations (\$482/PY) and outpatient visits (\$428/PY).



Psychiatric disorders (\$243/PY, 95% CI: \$226–\$249), contributing the largest share, exceeding the excess cost attributable to respiratory diseases (\$212/PY, 95% CI: \$208–\$217).